

Dirac Equation in T0 Theory: Introduction and Overview

Clifford Algebra, Spin Topology, and Geometric Integration

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Abstract

This document provides a brief introduction to the geometric interpretation of the Dirac equation within the framework of T0 theory. The Dirac equation is not fundamentally described by 4×4 matrices but by a Clifford algebra structure of spacetime. Spin-1/2 is a topological property (winding number on a torus), not a mysterious matrix property. In T0 theory, mass is determined dynamically by time-mass duality $T(x) \cdot m(x) = 1$, and the fractal dimension $D_f = 3 - \xi$ modifies the underlying metric.

For a complete technical presentation, see the main document: 051_dirac_En.pdf

.1 Overview

Integrating the Dirac equation into T0 theory requires a fundamental rethinking about the nature of Dirac matrices and spin. This brief document provides an overview of the most important concepts. For details, refer to the comprehensive technical document 051.

.2 The Fundamental Insight: Clifford Algebra

The Problem with 4×4 Matrices

The standard Dirac equation is usually written as:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \quad (1)$$

with complex 4×4 matrices γ^μ .

The question: Why 4×4 matrices? Are they fundamental?

The answer: No. The matrices are a **representation**, not the fundamental physics.

The Abstract Form

The fundamental Dirac equation is a Clifford algebra equation:

$$(i\mathbf{e}_\mu \partial^\mu - m)\Psi = 0 \quad (2)$$

where:

- $\mathbf{e}_\mu$: Abstract basis vectors of spacetime (not matrices!)
- Ψ : Geometric object in the spin bundle
- Clifford rule: $\mathbf{e}_\mu \mathbf{e}_\nu + \mathbf{e}_\nu \mathbf{e}_\mu = 2g_{\mu\nu}$

The 4×4 matrices γ^μ are only **one possible matrix representation** of the abstract basis vectors $\mathbf{e}^\mu$.

Representation vs. Physics

Fundamental: Clifford algebra structure

Representation: 4×4 matrices (calculation tool)

The matrices are **not** the physics, but a tool for calculation.

.3 Spin as Topology

The 720° Rotation

Spin-1/2 particles have the well-known property:

$$R(360^\circ)\Psi = -\Psi \quad \text{and} \quad R(720^\circ)\Psi = \Psi \quad (3)$$

This is **not a matrix property**, but follows directly from the Clifford algebra structure!

Winding Numbers on the Torus

In T0 theory, spin is interpreted geometrically:

$$\text{Spin-}s \longleftrightarrow \text{Winding } (n_\theta, n_\phi) \text{ with } \frac{n_\phi}{n_\theta} = 2s \quad (4)$$

Spin-1/2: Winding (1, 1) on the torus

The 720° rotation = traversing this winding twice

This is **pure topology**, not a mysterious quantum property!

.4 T0 Integration: Overview

Fractal Spacetime

T0 theory postulates a fractal spacetime dimension:

$$D_f = 3 - \xi \quad \text{with} \quad \xi = \frac{4}{3 \times 10^4} \quad (5)$$

This modifies the Clifford algebra structure to:

$$\mathbf{e}_\mu^{(\text{frak})} \mathbf{e}_\nu^{(\text{frak})} + \mathbf{e}_\nu^{(\text{frak})} \mathbf{e}_\mu^{(\text{frak})} = 2g_{\mu\nu}^{(\text{frak})} \quad (6)$$

Time-Mass Duality

Mass is not constant but dynamic:

$$T(x) \cdot m(x) = 1 \quad \Rightarrow \quad m(x) = \frac{1}{c^2 T(x)} \quad (7)$$

The T0 Dirac equation becomes:

$$(i\hat{\phi}_{\text{frak}} - m(x))\Psi(x) = 0 \quad (8)$$

Predictions

The fundamental prediction is a **ratio**:

$$\frac{a_\tau}{a_\mu} = \left(\frac{m_\tau}{m_\mu} \right)^2 \approx 283 \quad (9)$$

This is:

- Independent of unit systems
- Independent of fractal corrections
- Testable at Belle II (2027-2028)

.5 For Further Details

This brief overview covers only the most important concepts. For a complete technical presentation, see:

Main Document

Dirac Equation in T0 Theory: Geometric Integration

051_dirac_En.pdf

This document contains:

- Complete Clifford algebra formulation
- Detailed spin topology with figures
- Tetrad formalism for fractal metric
- Mass-proportional coupling and loop diagrams
- Time field dynamics in detail
- Natural vs. SI units
- Experimental tests and predictions
- Limits of the theory (honestly presented)

.6 Comparison Table

.7 Core Messages

1. The Dirac equation is fundamentally a **Clifford algebra equation**, not a matrix equation

Aspect	Standard Dirac	T0 Dirac
Mathematics	4×4 matrices	Clifford algebra
Spin	Matrix property	Topological winding
Mass	Constant m	Dynamic $m(x, t)$
Metric	Minkowski $\eta_{\mu\nu}$	Fractal $g_{\mu\nu}^{(\text{frak})}$
Dimension	$D = 4$	$D_f = 3 - \xi$ (space)
Topology	None	Torus
Predictions	Qualitative	Testable ratios

Table 1: Comparison: Standard vs. T0 Dirac formulation

2. Spin-1/2 is a **topological property** (winding number), not a mysterious matrix property
3. In T0 theory, mass is **dynamically** determined by time-mass duality
4. The fractal dimension modifies the **geometric structure** of spacetime
5. The testable prediction is the **ratio** $a_\tau/a_\mu = (m_\tau/m_\mu)^2$

Further Reading

T0 Theory Basics:

- Chapter 2: Xi-Narrative – Basic principles
- Chapter 3: Time-Mass Duality in QM and QFT
- Chapter 5: Predictions and Experimental Tests

Technical Details:

- 051_dirac_En.pdf – Complete Dirac integration

Clifford Algebras in General:

- Hestenes, D. "Space-Time Algebra"
- Lounesto, P. "Clifford Algebras and Spinors"
- Doran, C. & Lasenby, A. "Geometric Algebra for Physicists"